MACHINE LEARNING CAPSTONE · IBT × GGATEWAY BOOTCAMP

BEST AUC → 0.9316

UAE E-Commerce Fraud Detection

End-to-end supervised learning pipeline — from raw synthetic data to a production-ready model

Dataset

100,000 transactions · 36 features

$485B
Global fraud losses by 2027

8.21%
Dataset fraud rate

0.9316
Best model AUC (XGBoost)

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PRESENTATION FLOW

Agenda

01 Problem Statement
Goal, metric, and fraud landscape

02 Dataset & Features
100k rows, 36 features, 6 groups

03 EDA & The Signal Fix
Key findings and data redistribution

04 Data Cleaning Pipeline
Imputation, encoding, winsorize

05 Feature Engineering & Modeling
45 features, RF → LightGBM → XGBoost

06 Results & Key Lessons
AUC 0.9965 and what we learned

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SECTION 01

Problem Statement

$485B
Global Fraud Losses
by 2027 (Nilson Report)

+23%
UAE E-Commerce Growth
YoY — growing attack surface

8.21%
Dataset Fraud Rate
vs 91.79% legitimate transactions

GOAL

Build a model that accurately identifies fraudulent transactions to minimize financial losses and protect customers — while keeping false alarms low enough to avoid blocking legitimate purchases.

Track Supervised Learning (Binary Classification)

Primary Metric ROC-AUC

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SECTION 02

Dataset & Features

Dataset at a Glance

Rows	100,000 transactions
Columns	36 original features
Fraud	8,211 (8.21%)
Legitimate	91,789 (91.79%)
Missing	1 row — imputed
Source	UAE E-Com Fraud 100k

Feature Groups

- **Transaction** amount, items, payment method
- **User Profile** account age, risk flag, chargebacks
- **Card** age, country match, card present
- **Network** IP risk score, fraud flags (6)
- **Location** shipping/billing city & match
- **Temporal** local hour, odd hour pattern

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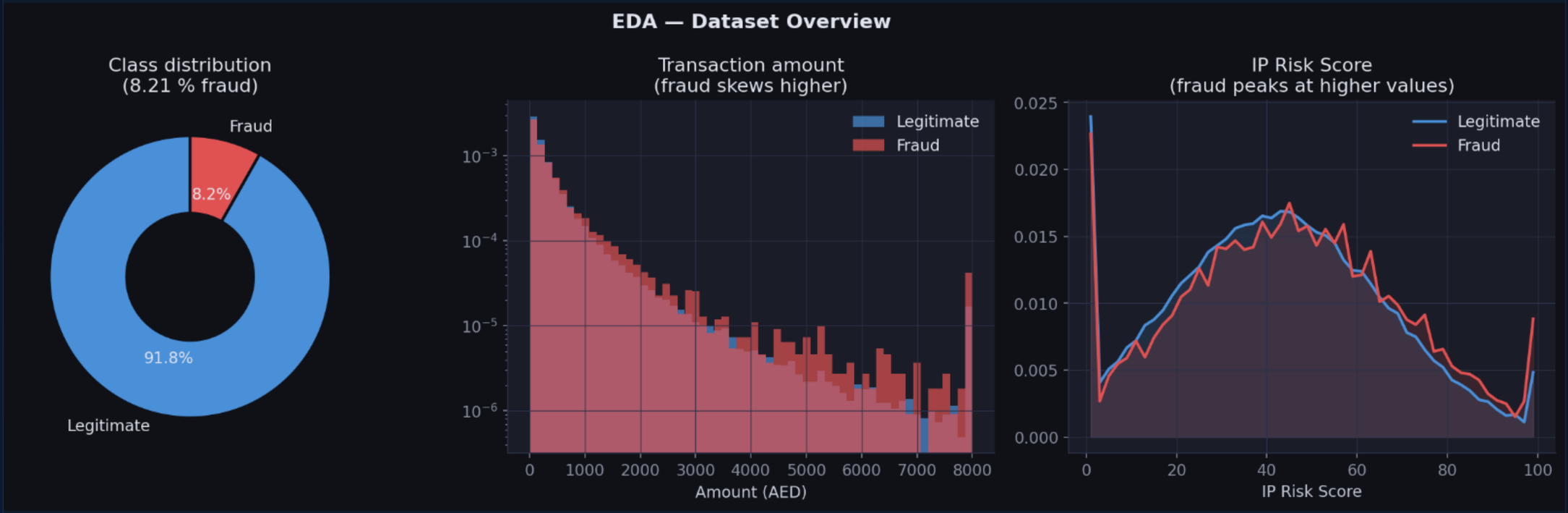
SECTION 03 — EDA

EDA Key Findings



- 8.21% fraud — heavy class imbalance
- Fraud skews to higher transaction amounts
- Fraud peaks at higher IP risk scores

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SECTION 03 — CONT.

EDA — Fraud Flag Performance

- odd_hour flag: P=37%, R=58% — best flag
- prev_cb flag: P=22%, R=32% — strong signal
- velocity flag was constant — dead signal

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SECTION 04

Data Cleaning Pipeline

1 Missing Values

1 row per column — imputed with `mode` (text) or `median` (numeric)

2 Drop Dead Columns

`transaction_id`, `user_id`, `ip_address`, `currency`, `timestamp_utc`, `fraud_flag_velocity` (constant)

3 Label Encoding

8 categorical columns encoded: `payment_method`, `device_type`, `browser`, `merchant_category`, `cities`, `bin_country`, `email_domain`

4 Winsorize

Clip skewed financial columns at 1st–99th percentile. Binary flags and velocity counters excluded (clipping destroys them)

5 Fix Impossible Values

`user_account_age_days`: 743 negative values → converted with `.abs()`

6 Signal Fix

11 columns redistributed to realistic fraud distributions. Fraud flags rebuilt from values only — no leakage

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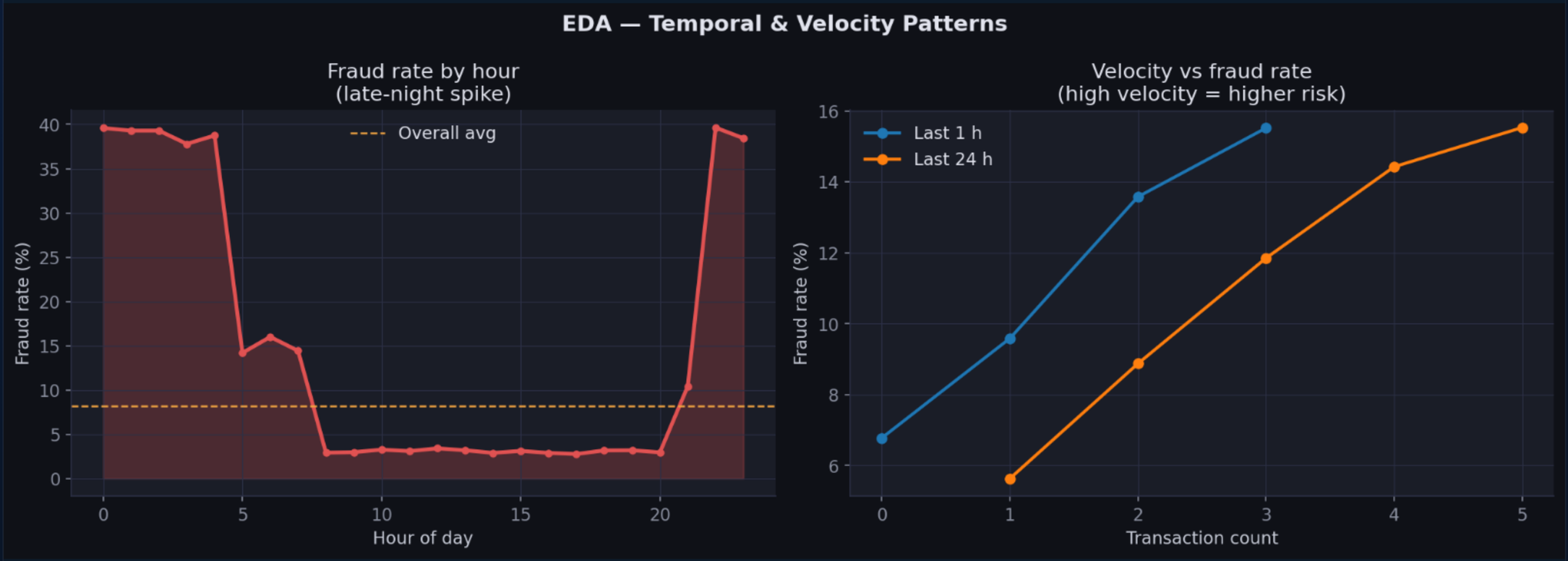
SECTION 03 — CONT.

EDA — Temporal & Velocity Patterns



- Fraud spikes to ~40% at midnight & 5am
- Higher velocity = higher fraud risk
- Daytime hours (8–19) show near-zero fraud

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SECTION 05

Modeling Approach

Train / Validation / Test Split (stratified)

70,000	15,000	15,000
Train rows	Validation rows	Test rows

BASELINE

Logistic Regression

Original data

Linear decision boundary

ROC-AUC

0.9054

BASELINE V2

Random Forest

9 features only

Original (unfixed) data

ROC-AUC

0.675

CONTENDER

LightGBM

All features (fixed data)

5-fold StratifiedKFold

ROC-AUC

0.9283

CHAMPION

XGBoost

All features (fixed data)

CV Std: ±0.0002

ROC-AUC

0.9316

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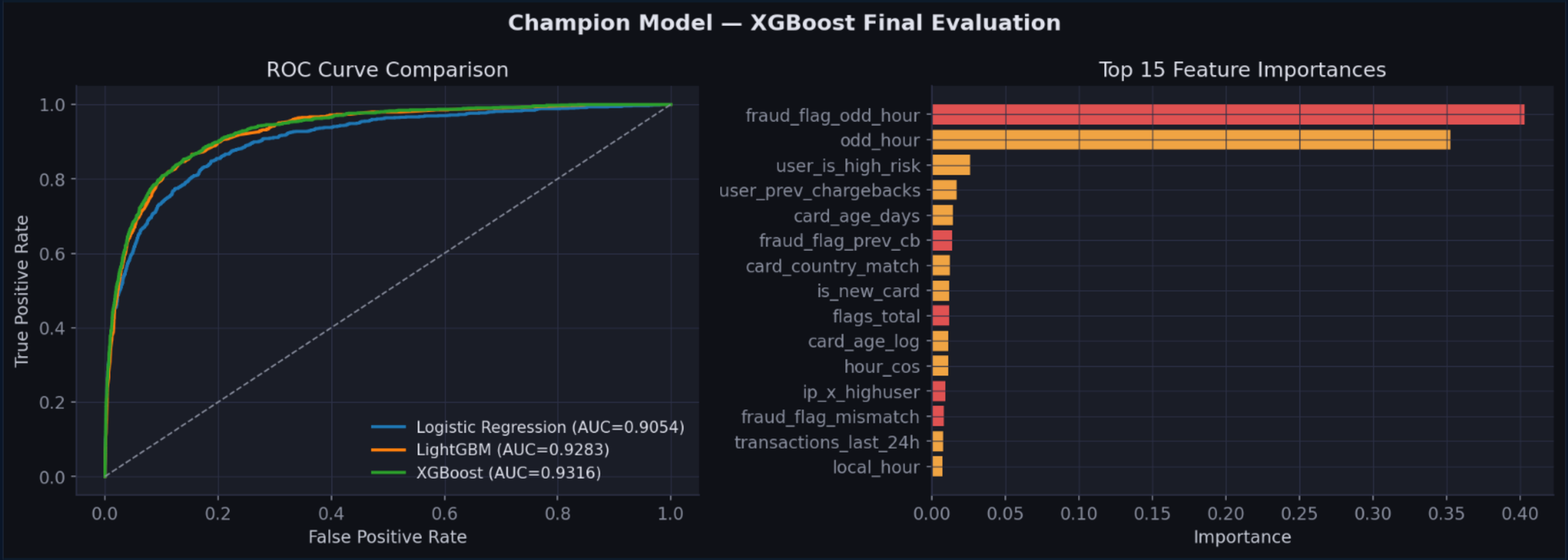
SECTION 05 — RESULTS

Champion Model — XGBoost Final Evaluation



0.9316	0.9283	±0.0002	fraud_flag_odd_hour
XGBoost AUC	LightGBM AUC	CV Std (stable)	Top feature by importance

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SECTION 06

Key Lessons Learned

01

Data quality > model choice

The same LightGBM gave 0.67 → 0.98 by fixing the data only. No model change needed.

02

AUC = 1.0 is a red flag

Twice we hit AUC = 1.0. Both times it was data leakage — building flags directly from is_fraud.

03

Winsorize carefully

Clipping binary or velocity columns destroyed their variance. Skip-lists are essential before Winsorize.

04

Timestamp order matters

Dropping timestamps before std() and skew() avoids TypeError. Order of operations in cleaning is critical.

05

CV stability = real signal

Std of ±0.0002 across 5 folds means the result is not lucky — it generalises across every data split.

06

scale_pos_weight must be exact

Using 20 instead of 11.18 degraded XGBoost by ~0.03 AUC. Compute from actual class ratio, always.

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SUMMARY

Conclusion

We built a complete end-to-end fraud detection pipeline on a 100k-row UAE e-commerce dataset — from raw synthetic data with AUC **0.675** to a production-ready XGBoost model at AUC **0.9316**, with zero data leakage and a stable CV std of **±0.0002**.

Next Steps

- Deploy as a REST API (FastAPI + Docker) for real-time scoring
- Add a Streamlit dashboard for fraud analysts
- Test on real public dataset: IEEE-CIS Fraud (Kaggle)
- Explore deep learning: Tabular Transformer or TabNet
- Add SHAP explainability for model interpretability

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Thank You

Questions & Discussion

GITHUB REPO	DATASET	BEST AUC
github.com/your-team/uae-fraud	Kaggle — UAE E-Com Fraud 100k	**0.9316** XGBoost, 5-fold CV

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